

Gwon Ryul Han

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Education

Ulsan National Institute of Science and Technology (UNIST)
B.S. in Physics

Ulsan, Republic of Korea
March 2024 – Present

Research Interests

- Quantum Information Theory, Quantum Metrology, Quantum State over Spacetime
- Mixed State Topological Order, Anyon Theory

Publications

- [1] Gwon Ryul Han, Hyukgun Kwon, and Seok Hyung Lie. *Quantum Metrology for Signals with an Unknown Causal Structure*. 2026. arXiv: 2609.28066 [quant-ph]. URL: <https://arxiv.org/abs/2609.28066>.

section*Publication

Research Experience

Quantum Information Theory Group

Undergraduate Researcher(Advised by Seok Hyung Lie)

UNIST June 2024 – Present

- **Research Topic:** Quantum Metrology, Quantum State over Spacetime

Topological Matter Theory Group

Undergraduate Researcher(Advised by Gilyoung Cho)

KAIST June 2026 – Present

- **Research Topic:** Mixed State Topological Order, Anyon Theory (related to Quantum Error Correction Code)